Muhammad Mohsin

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Summary

Senior full-stack engineer with 6+ years building production systems in medtech and fintech, from embedded hardware to AWS. At Dosell, built the OCR pipeline that auto-resolves about 680,000 dispenser alarms a year at 99.9% (13x faster, saving about USD 18,000 a year) and led the Flutter app behind a 1M SEK ARR milestone. Previously built a Bitcoin ATM network of 22 machines across Spain, end to end from device protocols to backend. Also builds and runs independent products: DriveTest.se, Drive in Sweden and jobflow. Works across Python/Django, TypeScript/Node, Flutter, AWS and Terraform, owning systems from design to production.

Experience

Dosell AB — Stockholm, Sweden

Full Stack Engineer — *Dec 2022 - Present*

- Built a parallel OCR pipeline (Django, Celery) on AWS ECS, deployed as Terraform/Terragrunt IaC, auto-resolving about 680,000 dispenser alarms a year across 3,972 devices at 99.9%, cutting handling from 10+ minutes to 47 s per alarm (13x faster) and saving about USD 18,000 a year.
- Cut care-staff portal response times from 10-15 s to 1-3 s (5-10x faster) by adding read-replica routing, rewriting alarm queries, archiving and indexing audit events, and removing hidden outbound API calls in the NestJS/DocumentDB backend.
- Shipped Microsoft Entra ID SSO end to end (MSAL backend, group-to-role mapping, management UI), unlocking hospital-scale enterprise customers.
- Designed and built two products from scratch to production: WebLoad (auth, load/unload/merge flows, QR scanning, push notifications) and the Dosell Pro Flutter tablet app (patient UI, video calls, supervised dispense, multi-language).
- Engineered a production data lake (AWS DMS to S3 Parquet, Glue ETL decryption job, Athena) making 48.9 GiB queryable via SQL, plus daily Lambda FTP/SFTP crawlers archiving 365+ GiB of device files.
- Led development of Dosell's first hybrid application (Flutter, iOS/Android/Web) within a 2-month deadline, enabling a Finnish deal of 500-1000 units and securing the 1M SEK ARR milestone.
- Designed a microservices architecture with NestJS, PostgreSQL and event-driven patterns to replace legacy systems, increasing throughput 3x and reducing latency from 2000 ms to 300 ms.
- Architected React.js/Next.js medical applications with code splitting and responsive design, improving load times by 40% and achieving 99.9% uptime.
- Mentored five junior developers through code reviews, architecture guidance and workshops on React, Node.js and cloud deployment; all were promoted within 12 months.

Matrix Network International B.V. — Remote

Software Engineer — *Apr 2020 - Nov 2022*

- Architected a complete Bitcoin ATM solution fully remotely (hardware integration, AWS SQS microservices backend, website, Bitcoin mainnet/Lightning Network), deploying 22 ATMs across Spain for secure cash and debit-card purchases.
- Built the Venrecarga mobile recharge service with a multi-provider payment gateway (bunq, Stripe, PayPal, OpenNode), handling high-volume top-ups for the Venezuelan diaspora worldwide during the economic crisis.
- Developed the Rembit crypto remittance service over Mastercard and Visa networks with KYC and AML checks and a 24-hour automated S3-to-MySQL ETL pipeline via Athena, processing EUR 10,000+ in Bitcoin transfers in its first month.

Projects

jobflow | *Creator (web app + browser extension)*

- Built a free, private job-hunt platform that finds matching jobs, tailors the CV for each one without inventing anything, tracks replies from the inbox and fills in application forms via a Chrome extension.
- Stack: Django, PostgreSQL, Gemini AI, Gmail integration and a Chrome extension.

DriveTest.se | *Creator (website + mobile app)* 2023 - Present

- Built an exam-prep platform for the Swedish taxi, transport and driving licence theory tests with 1000+ exam-style questions, an AI tutor that explains every answer, multilingual questions and progress tracking.
- Ships on Android and the web (Flutter, Django REST) with a bilingual Next.js/TypeScript website of SEO exam guides.

Drive in Sweden | *Creator (web app + SaaS)*

- Built a SaaS that gets learner drivers an earlier driving test by watching for free test times around the clock and alerting users by email and notification the moment one opens nearby, so they can book in a few clicks.
- Stack: Cloudflare Workers, Hono, D1, Next.js, TypeScript, Stripe payments and BankID login.

Bitcoin ATM (ATMbitbit) | *Full Stack Developer (hardware + payments)* 2020 - 2022

- Built a Bitcoin ATM end to end: insert cash or use a debit card and receive Bitcoin instantly over the Lightning Network, with hardware integration and payment processing; 22 machines deployed across Spain.

CamBridge | *Creator (Swift, iOS + macOS)*

- Built native iOS and macOS apps that turn an iPhone into a pro camera watched and controlled from a Mac or a second iPhone: photos, video, lens switching, zoom, focus and exposure, in portrait or landscape.
- Built with Swift, SwiftUI, AVFoundation, VideoToolbox and Network.framework for creators who don't want to buy a camera rig.

Skills

Python: Django, Django REST Framework, Celery, pandas

JavaScript/TypeScript: Node.js, NestJS, Express.js, React.js, Next.js

Mobile: Flutter, Dart

AWS & Infrastructure: Terraform, S3, EC2, Lambda, RDS, DocumentDB, DynamoDB, SQS, Athena, CloudWatch, CloudFront, Docker, CI/CD, Linux

Databases: PostgreSQL, MySQL, SQL Server, Oracle DB, MongoDB, DynamoDB, Firestore

System Design: Microservices, event-driven architecture, Kafka, RabbitMQ, Elasticsearch/OpenSearch, Airbyte, ETL pipelines

Hardware: ESP32/ESP8266, Raspberry Pi, MDB protocol, bill & coin validators, serial communication, vending machines

AI Tools: Claude Code, GitHub Copilot, OpenAI, Gemini, Ollama, n8n

Awards & Achievements

– Oracle Certified Java Master Trainer

– Oracle Certified Java Foundations

Education

Blekinge Institute of Technology Karlskrona, Sweden
MS Computer Science *2022 - Present*

University of Gujrat Gujrat, Pakistan
BS Computer Science *2016 - 2020*